

Task 3 Report: Post-Deployment Monitoring of SCB-CS-001

This report uses the Task 3 notebook outputs to assess whether the SCB-CS-001 retail credit scoring model remains reliable after deployment through RegLens AI, the proposed RegTech monitoring prototype. The analysis compares the 2022-2023 training data with the 2025-2026 live data, then interprets the same evidence through OCC and MAS compliance lenses.

1. Monitoring Question

The central question is not simply whether the model still produces scores, but whether those scores are being used in a population that still resembles the training population. For a credit model, this matters because a shifted applicant base can change both portfolio risk and the fairness profile of approval decisions.

2. Data and Method

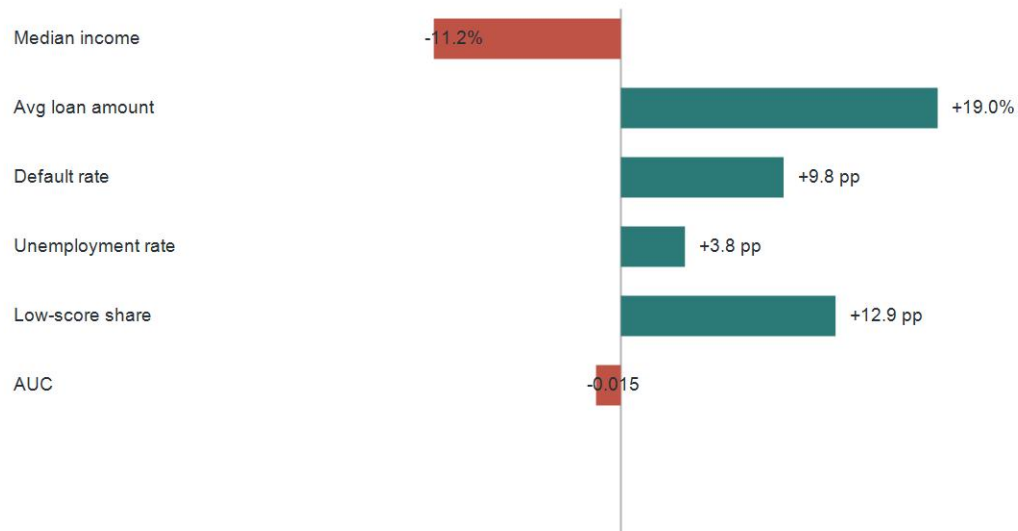
RegLens AI compares 5,000 training-period applications and 3,000 live-period applications. Both files contain applicant characteristics, demographic fields, model score, approval outcome and actual default outcome. The monitoring method uses business drift metrics, PSI, AUC and subgroup approval rates.

Metric	Training	Live	Change
Median income	SGD 47,832	SGD 42,454	-11.2%
Average loan amount	SGD 77,580	SGD 92,349	+19.0%
Approval rate	98.48%	90.00%	-8.48 pp
Default rate	24.86%	34.63%	+9.77 pp
Unemployment rate	2.82%	6.67%	+3.85 pp
Low-score share	3.24%	16.13%	+12.89 pp
AUC	0.6312	0.6162	-0.0150

3. Portfolio Drift

The live portfolio is visibly weaker than the training-period portfolio. Median income is lower, average loan size is higher, unemployment is higher, and realised default has increased. This does not prove that the model is unfair, but it does show that the deployment environment has changed.

Training-to-Live Change in Key Monitoring Indicators



Negative values indicate deterioration where relevant; pp = percentage points.

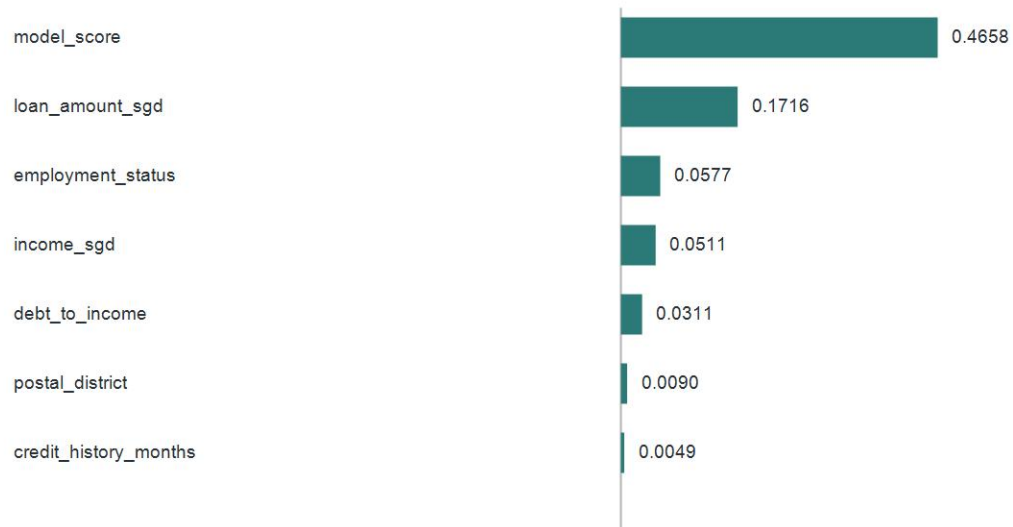
4. Distribution Shift Using PSI

PSI gives a more formal view of distribution shift. The strongest signal is the model score distribution. A model-score PSI above 0.25 is normally treated as significant drift, so this is the clearest trigger for model review.

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model_score	0.4658	Significant drift
loan_amount_sgd	0.1716	Moderate drift
employment_status	0.0577	Low / no material drift
income_sgd	0.0511	Low / no material drift
debt_to_income	0.0311	Low / no material drift
postal_district	0.0090	Low / no material drift
credit_history_months	0.0049	Low / no material drift

gender	0.0001	Low / no material drift
ethnicity	0.0001	Low / no material drift

Population Stability Index by Variable



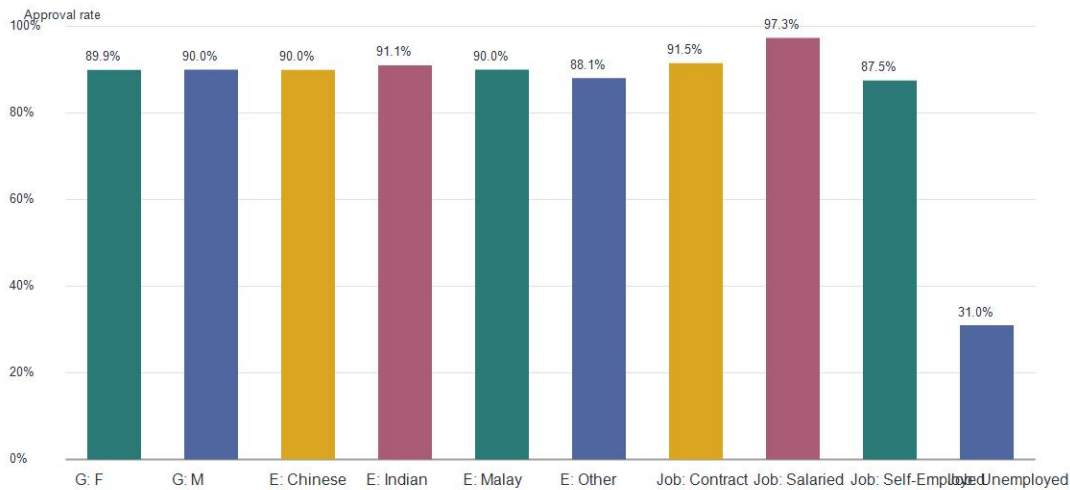
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5. Performance and Subgroup Outcomes

AUC falls from 0.6312 to 0.6162, which indicates weaker separation between non-default and default cases in live deployment. The decline is not dramatic by itself, but it is consistent with the broader drift pattern.

Gender and ethnicity approval rates are relatively close in the live data, so the evidence does not support a simple conclusion of direct protected-class approval disparity. The much larger gap appears by employment status. That gap may be credit-risk related, but it still requires explanation testing and proxy-risk review.

Live Approval Rates by Subgroup



6. Jurisdiction-Specific Interpretation

Jurisdiction	Primary concern	Interpretation from this dataset	Prototype output
OCC / United States	Fair lending, adverse action, consumer harm	Gender and ethnicity approval gaps are small, but employment-status gap and deteriorating performance justify review.	Amber: fair-lending and adverse action review
MAS / Singapore	AI governance, model drift, validation, oversight	Model score PSI is above the significant-drift threshold and live default risk is materially higher.	Red: governance escalation and validation review

The difference between OCC and MAS is therefore a difference in emphasis. The OCC lens asks whether the model creates unfair lending outcomes or weak adverse action explanations. The MAS lens asks whether the bank is governing a changed model environment responsibly. In this dataset, the OCC result is best framed as an amber review, while the MAS result is closer to a red governance escalation.

7. Recommendation

RegLens AI's recommendation is that the bank should not continue relying on the original validation evidence alone. A refreshed validation exercise should be performed, including PSI monitoring, AUC review, score-band default analysis, subgroup approval monitoring and

documentation of adverse action explanations. The model may not need to be withdrawn immediately, but it should be recalibrated or benchmarked against a challenger model before continued large-scale use.

References

Office of the Comptroller of the Currency. Model Risk Management: Revised Guidance, OCC Bulletin 2026-13.

Consumer Financial Protection Bureau. Guidance on credit denials by lenders using artificial intelligence.

Monetary Authority of Singapore. FEAT Principles and Responsible AI/Data Analytics guidance.